

ANYA JHA

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Experience

HCLTech | Software Developer, Client: PayPal

October 2025 – Present
Chennai

Agentic Commerce — *Feature Development, Test Coverage & Continuous Integration Reliability*

- Built the delegated merchant-access feature end-to-end in **Next.js**, **React**, and **TypeScript**, wiring the assignment interface to a live engineer directory and fixing a missing security token that was blocking every submission, giving support engineers self-service control over merchant access without manual backend intervention.
- Designed and built a **Cypress** suite in **TypeScript** covering configuration management, bulk data upload, and delegated access control, increasing functional test coverage from **zero to 100%** of these critical flows.
- Authored **Jest** suites against uncovered branches in the merchant-access module and replaced **Cypress**'s open-ended retries with bounded, deterministic polling, raising new-code test coverage from **26% to 96.7%** (21 suites, 225 tests) and cutting the continuous integration failure rate from **19% to under 5%**.
- Corrected **SonarQube**'s coverage measurement scope at the pipeline level to stop compiled build output being counted as untested source code, closing **4** coverage gaps it had been masking.

IBM | Digital Technical Specialist (Summer Intern)

June 2024 – Aug 2024
Bengaluru

Observability Benchmarking & AI Agent Design

- Benchmarked **4 enterprise observability platforms** across **10+ parameters**, identified **3 capability gaps** in IBM Instana's incident response layer – findings accepted by the Director of the Digital Technical Specialist team and flagged for the **product roadmap**.
- Authored a technical spec for an autonomous AI remediation agent targeting a **35–40% reduction in mean time to resolution**, defining system architecture for real-time resolution suggestions at point-of-failure.
- Supported **15+ enterprise prospect evaluations** through live technical demos of IBM Instana, contributing to client conversions within a **2-month** window.

Projects

Deep Packet Inspection Engine [GitHub]

C++, CMake, Python

- Built a multi-threaded DPI engine in **C++** that parses PCAP traffic, extracts domains via **TLS/SNI inspection**, and enforces application-level blocking rules across **5-tuple flows**.
- Designed a concurrent pipeline with **consistent hashing** across worker threads for flow-aware packet distribution, enabling stateful connection tracking without locks on shared flow state.
- Implemented a configurable rule engine supporting **IP, domain, and application-level** blocking with per-application traffic statistics and filtered PCAP output generation.

Technical Skills

Languages: Java, JavaScript, TypeScript, Python, SQL

Frameworks & Libraries: React, Next.js, Node.js

Tools & Infrastructure: Git, Docker, AWS, Postman, REST APIs, Harness CI/CD, SonarQube, Cypress

Artificial Intelligence: RAG Architectures, Embedded Systems, LangGraph

CS Fundamentals: Data Structures & Algorithms, System Design, Object-Oriented Programming

Education

Indian Institute of Technology

Patna, Bihar
2026

Master of Technology, Artificial Intelligence and Data Science

DIT University

Dehradun, Uttarakhand
July 2021 – June 2025

Bachelor of Technology, Computer Science and Engineering | **CGPA: 9.13/10**

Achievements

Emcee, PIARC International Conference, France | National-level Debate Competitor | State-level Athlete, Track & Field